Where Does Vision Belong in a Frozen Time-Series Forecaster?

1 Settings

All metrics are normalised by installed capacity, so they are comparable across systems of different size, and all are computed on daytime steps only.

Quantity / Metric	Specification / Description
Installations	100 UK residential rooftops, 1.5–4.0 kW; two carry a data-quality flag and are dropped, leaving 98
Sampling	every 30 minutes, years 2019 and 2020
History window	14 days = 672 samples of the plant's own power
Forecast horizon	6 hours = 12 samples, 9 quantiles emitted per step
Covariates	*Weather*: temperature, cloud cover, wind speed, precipitation, radiation, irradiance. *Solar geometry*: solar zenith, solar azimuth, day-of-year, solar time, clear-sky GHI
Visual signal	8 satellite frames at 224 × 224 covering the 6 hours ending at the origin, centred on the installation
Split	70/15/15 **by installation**, not by time: train, validation and test share the same two years and share no rooftop
Evaluation set	14 test installations, 165,295 scored daytime steps
Seeds	42, 43, 44
Skill score (SS, ↑)	$\mathrm{SS} = 1 - \mathrm{NRMSE}_{\mathrm{model}}/\mathrm{NRMSE}_{\mathrm{ref}}$ against Smart Persistence ($\mathrm{SS} = 0$). Higher is better; one means perfection.
Ramp NMAE (↓)	NMAE restricted to steps where true power changes sharply between consecutive samples — a cloud edge crossing the array (lower is better).

Table 1: The experimental setting and evaluated metrics. All configurations in this report share this setting.

Smart Persistence is the field's standard null hypothesis and it is a strong one: on clear or uniformly overcast days it is nearly unbeatable.

Ramps deserve the same emphasis. They are a small minority of steps, but they carry nearly all of the operational cost of a forecast error.

2 The research question

The multimodal forecasting literature is largely a literature of proposals: an architecture is introduced, it is trained end to end, it beats a set of unimodal baselines, and the improvement is attributed to the additional modality. That attribution is rarely tested. When both encoders are trained jointly, an accuracy gain is consistent with at least three explanations — the model learned to read the images, the extra capacity helped, or the extra gradient path acted as regularisation.

This work inverts the emphasis. Both encoders are **pretrained and frozen**, so neither can adapt to the other; only a small bridge between them is learned. The architecture is then held fixed and the **attachment point** is varied. The question becomes:

> *Given a frozen sequence forecaster and a frozen visual encoder, at which point in the forecaster's computation must the visual representation be introduced before the forecaster demonstrably uses it?*

Freezing is what makes the question answerable. With both backbones fixed, no configuration can win by having more capacity than another, and any measured difference is attributable to the wiring. It is also what makes the result portable: the finding is a statement about placement in a frozen pipeline, not about one particular set of weights.

3 Components

Component	Role and dimensions	State
Chronos-2	Pretrained encoder-only time-series transformer.	frozen, top 3 of 12 blocks unfrozen
V-JEPA 2	Self-supervised video encoder. Encodes the 8-frame clip once, offline, into 4 temporal slices × 196 spatial patches × 1024 channels, cached.	frozen entirely
Channel projection	Maps the visual channel width onto the backbone's width.	trained
Fusion module	The only thing that differs between configurations.	trained

Table 2:

3.1 The forecasting backbone

The backbone treats a series the way a transformer treats any sequence: it is cut into non-overlapping patches of 16 consecutive samples, each patch is linearly projected into a token, and a 672-step history becomes 42 context tokens. Future positions are represented by additional placeholder tokens carrying only their known covariates; self-attention over the whole assembly lets those future positions absorb information from the past, and a projection head reads the answer off them.

The backbone is genuinely general-purpose: it was pretrained on heterogeneous series and has no notion of irradiance, panel tilt or clear-sky curves, so nothing about the solar domain is baked in. It also consumes covariates natively, which means the numeric-only control is already a strong model and the visual signal must earn its place against a high bar.

3.2 The visual encoder

The choice of a self-supervised video model rather than a caption-aligned one is deliberate. The relevant visual content — the motion and deformation of cloud fields — is temporal and has no useful textual description; a representation trained to match captions would discard exactly the structure that matters. V-JEPA's predictive objective in latent space is, by construction, a model of **how a scene changes**, which is the property the forecaster needs. The encoder is never run during training.

4 The four configurations

The three fusion arms are best understood not as three architectures but as three answers to a single structural question: **along which axis does the visual information enter?**

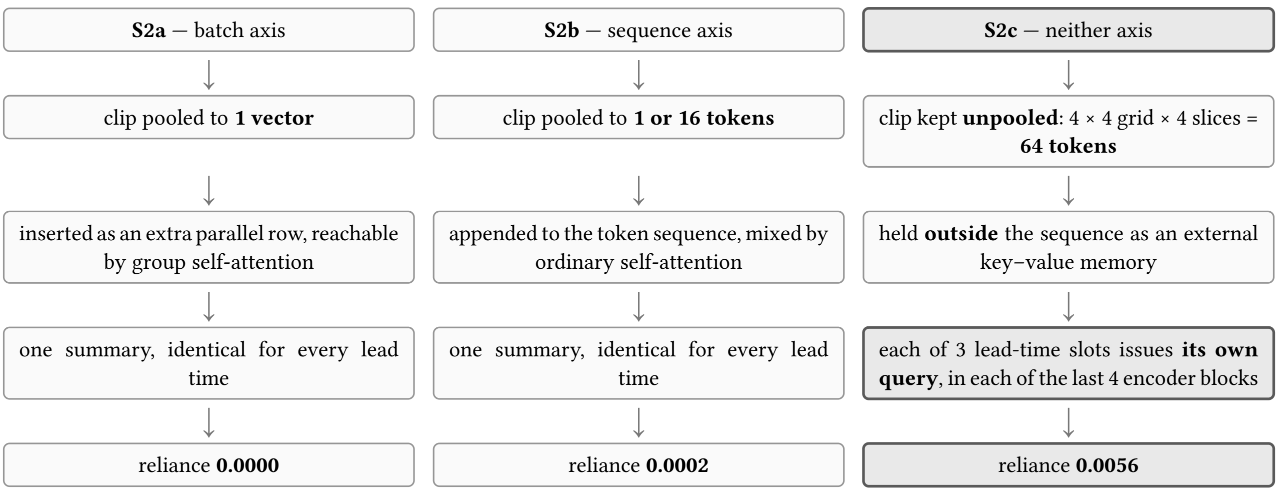

Figure 1:

4.1 S1 — the vision-free control

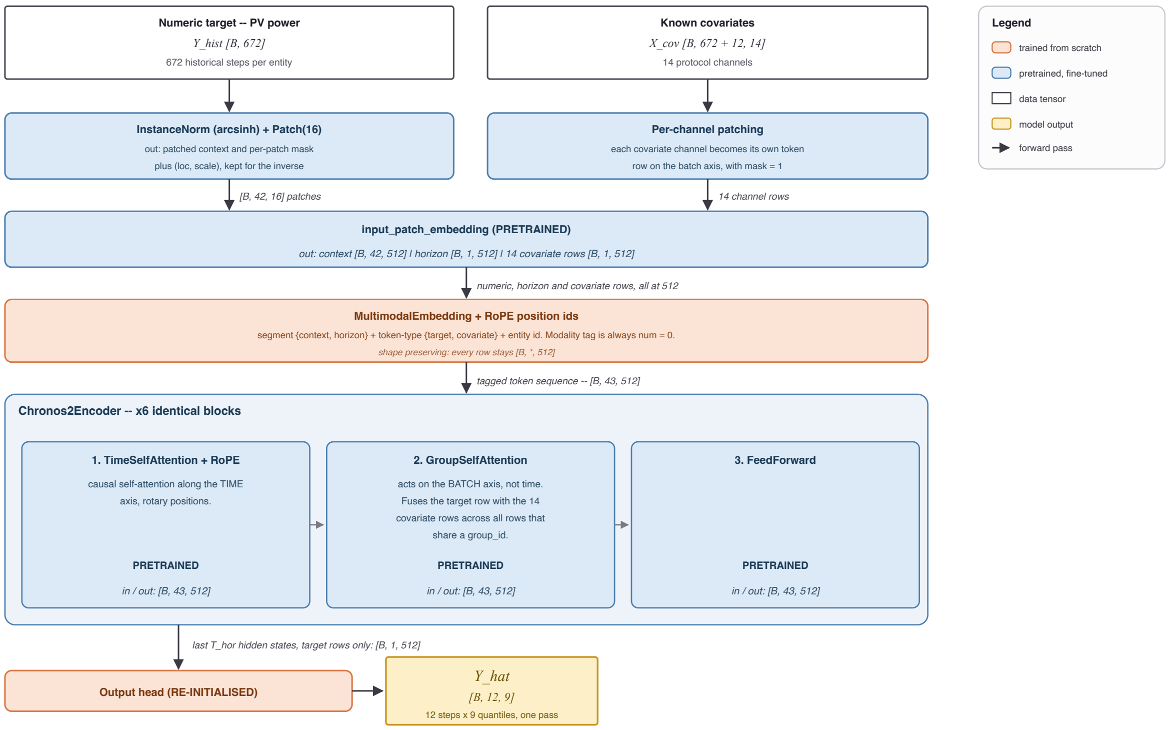

Figure 2: S1, the numeric-only control.

S1 is the same backbone with no imagery at all. It exists to define the baseline against which reliance is measured, ensuring that a multimodal model's advantage came from the fine-tuning recipe rather than from the second modality. Every subsequent arm is initialised from this checkpoint and trained with the same schedule, so any difference is attributable to the visual pathway.

4.2 S2a — fusion on the batch axis

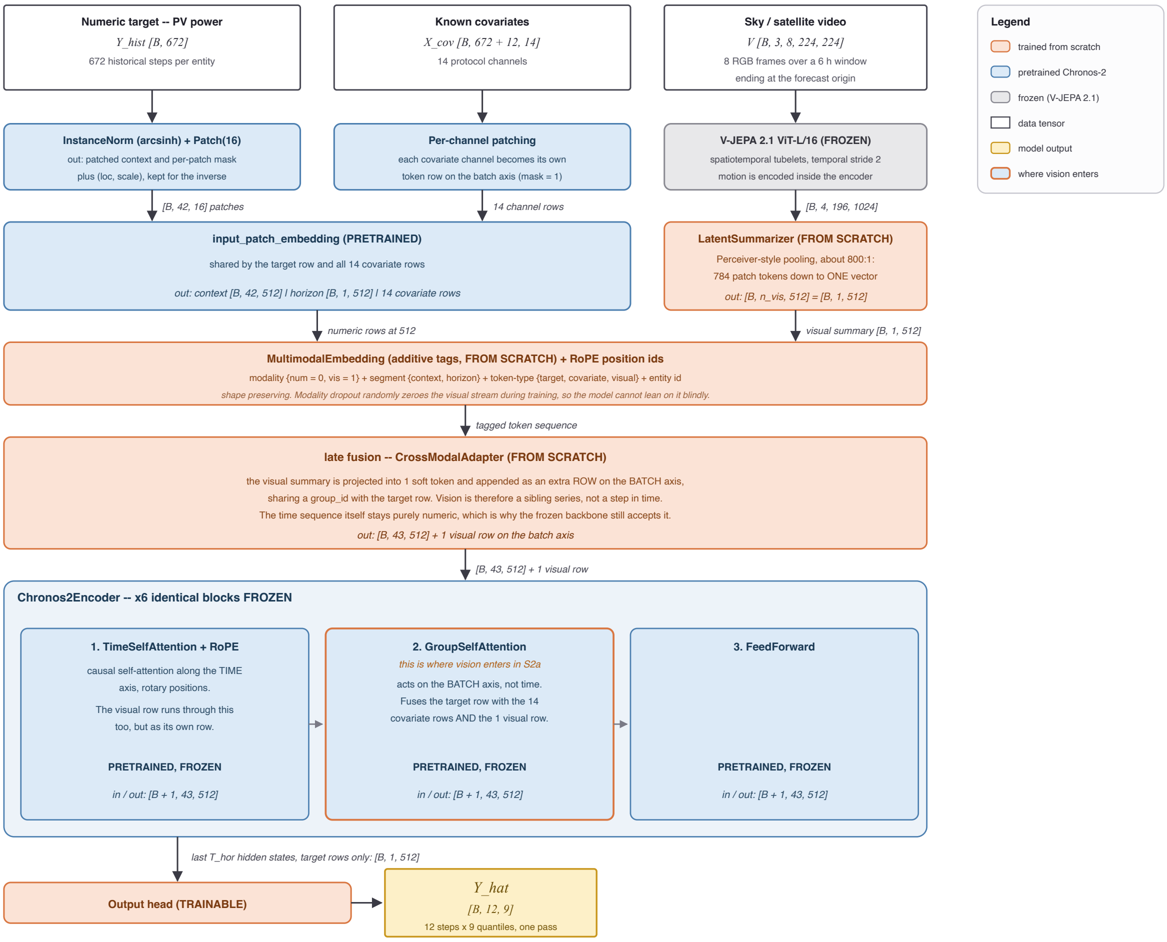

Figure 3: S2a, pooled late fusion.

The clip is pooled to a single descriptor and presented to the backbone as an extra parallel channel — a summary of “what today’s sky is like”, available to every forecast position equally. This is textbook late fusion, and it was the first arm precisely **because** it is the textbook answer: if the standard method works, the thesis is finished, and if it does not, the null is a result rather than an omission.

The rationale for expecting it to work is reasonable. The most obvious use of a satellite view is as a coarse regime indicator — clear, broken, overcast — and a single vector is an adequate carrier for a regime label. The rationale for expecting it to fail is equally clear in hindsight: the pooling operation compresses roughly 800 visual elements into 1, and it does so **before anything has been asked of them**. The summary must be computed without knowing which part of the scene is relevant, and by the time the forecaster is in a position to have an opinion, the discarded detail is gone.

4.3 S2b — fusion on the sequence axis

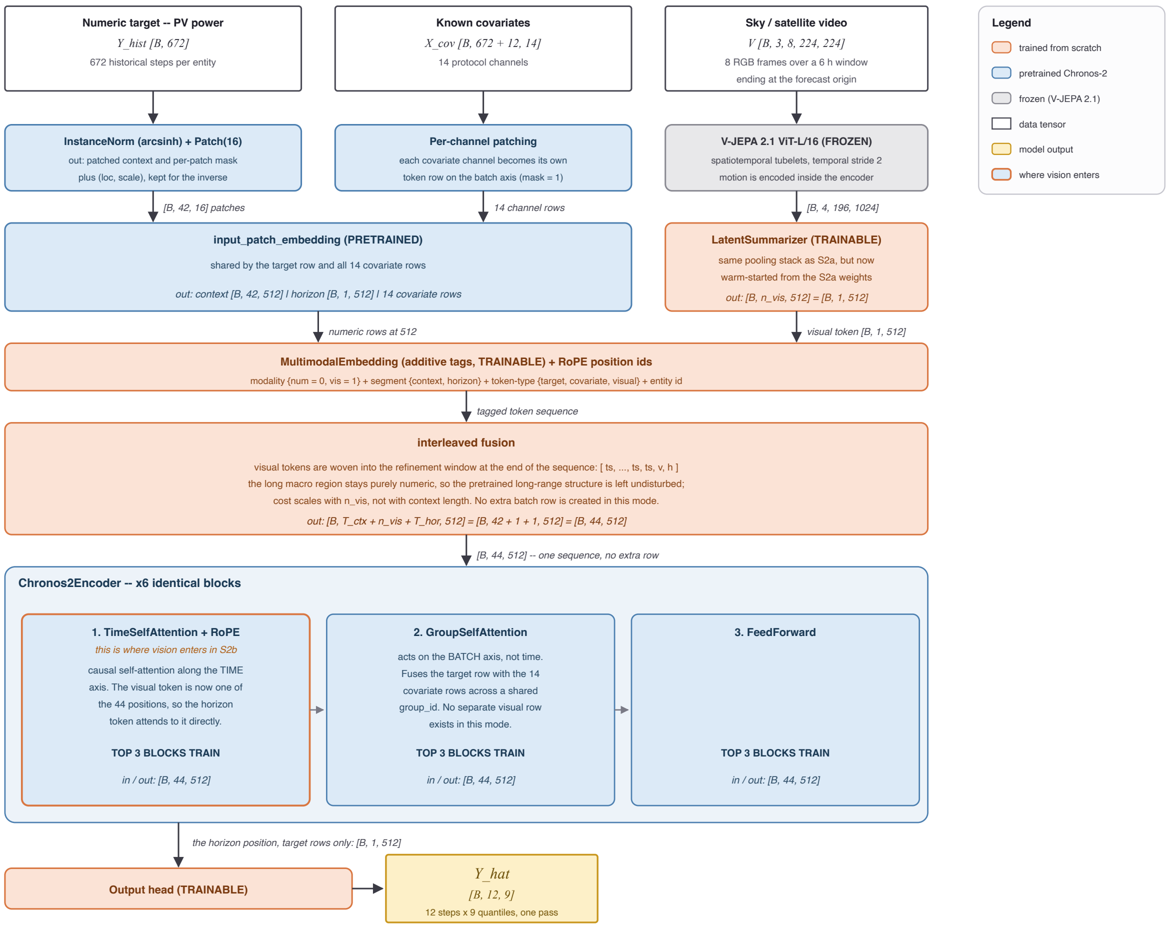

Figure 4: S2b, mid-sequence injection.

The natural next move is to stop treating vision as a parallel channel and make it part of the sequence, so that ordinary self-attention can relate visual tokens to numeric ones directly. Two widths were run — one visual token, and sixteen — to test whether the failure of S2a was simply a matter of bandwidth.

The reasoning is sound and the result is instructive. Widening the channel from 1 token to 16 did not help; the reliance stayed at zero within noise in both settings. That comparison is what rules out the bandwidth explanation and points at something structural: the problem is not **how much** visual information is admitted but **when** it is summarised relative to when it is needed. In both S2a and S2b the imagery is condensed into a fixed representation before the model has formed any view about the forecast, and every forecast position then receives the same representation.

4.4 S2c — the forecast queries the sky

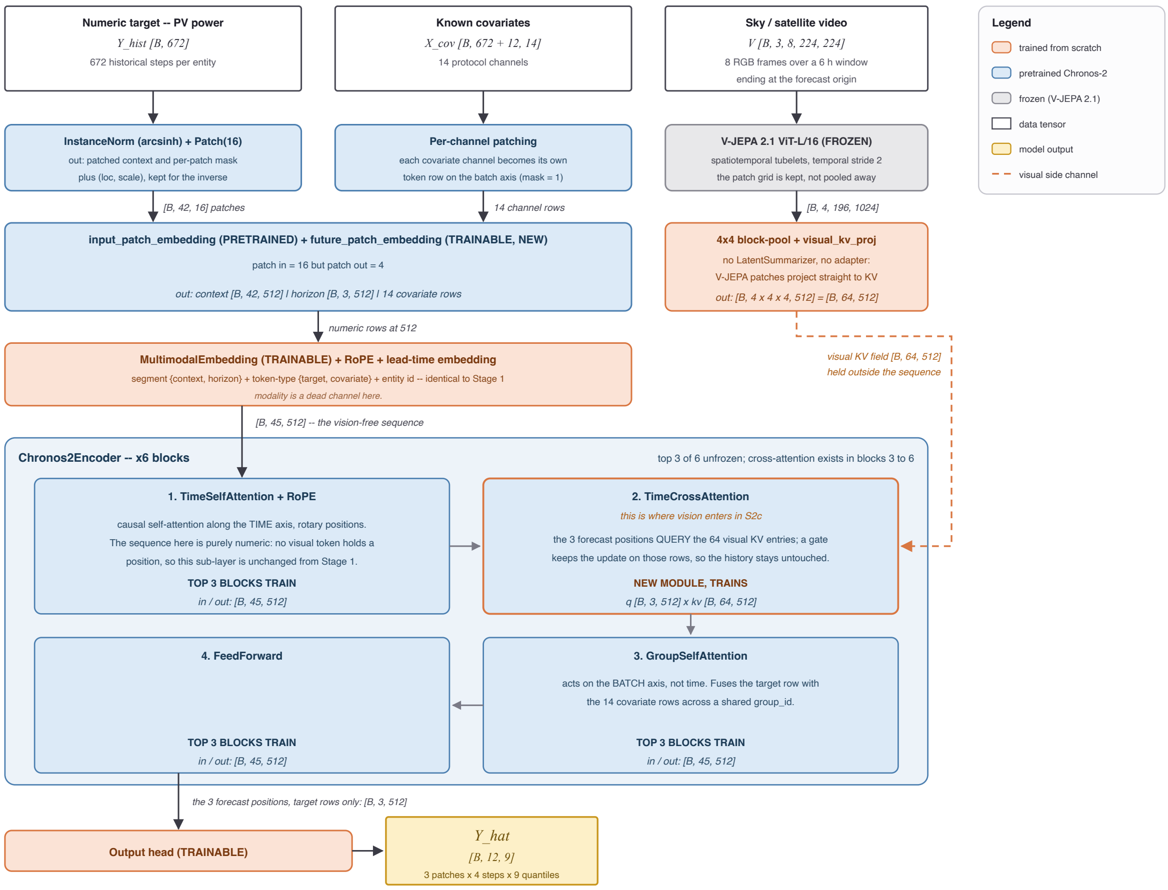

Figure 5: S2c, future-position cross-attention.

S2c changes the direction of the operation. The imagery is not inserted anywhere. It is retained **unpooled** as an external memory — the 14 × 14 patch grid is block-pooled to 4 × 4 and all four temporal slices are kept, giving 64 key–value tokens — and the forecast horizon is subdivided into three lead-time slots. Each slot issues its own cross-attention query against that memory, in each of the backbone's last four encoder blocks. No summarisation happens anywhere: the pooled descriptor and the projection adapter used by the other arms are both bypassed.

4.4.1 What a query actually is

A query is not made of the thing that is being predicted. It is a **description of what is being looked for** — the same object as a search string, which exists before the document it retrieves. Each forecast slot's query is assembled from exactly two ingredients, both available before any forecast value exists: a learned lead-time identity, a vector attached permanently to "the slot two hours out", shared across all samples and installations and fixed after training; and the slot's own hidden state after self-attention over the history and the known covariates, which encodes a **belief** about what is likely to happen, not the outcome.

self-attention populates the slot from history and covariates → slot forms a **query**: lead-time identity + its own belief → query reads the **64 unpooled visual tokens** → result **gated back** onto that slot alone → projection head emits the number

Figure 6: Order of operations for one lead-time slot in S2c, repeated in each of the last four encoder blocks. Each pass refines the slot's question with what the previous round returned.

4.4.2 Why the slots ask different things

Each lead-time slot receives gradient only from its own horizon's errors. The two-hour slot is penalised for two-hour mistakes, the five-hour slot for five-hour mistakes, and cloud fields relevant at those two horizons sit at different distances from the array. Nothing in the design instructs a slot to attend to cloud edges; the differentiation is the accumulated residue of one horizon's own past errors.

4.4.3 Why this is not late fusion, and not simply another cross-attention model

Why this is not late fusion. Conventional late fusion typically compresses an auxiliary modality into a shared representation before combining it with the time-series state. This creates a potential bottleneck: the same visual summary must support predictions at different future horizons, even though different horizons may require different information from the scene.

Why this is not simply another cross-attention model. Cross-attention itself is not the contribution. What matters is where the queries originate and what they retrieve. Existing multimodal forecasting approaches typically use historical time-series representations to query auxiliary features, enriching the context before a separate prediction stage. Here, instead, future lead-time slots query an unpooled visual memory directly. Each horizon can therefore retrieve different visual information relevant to its prediction.

5 The measurement instrument

After training, the model is frozen and the test set is evaluated twice: once normally, and once with the visual query switched off. The difference between the two passes is the **reliance** — the amount of the model's accuracy that depends on it having seen the sky.

This is a stronger instrument than comparing a multimodal model to a unimodal one, because it holds every weight fixed. It cannot be confounded by capacity, initialisation, or training schedule; the two passes differ only in whether the imagery was available.

6 Results

6.1 The placement result

Table 3 is the central finding. The reliance column reports the ramp-error improvement attributable to the imagery, measured by the counterfactual pass.

Arm	Where vision enters	Reliance (ramp)
S2a	batch axis, pooled	0.0000 ± 0.0015
S2b	sequence axis, 1 token	0.0006
S2b wide	sequence axis, 16 tokens	0.0002 ± 0.0016
S2c	**future-position queries**	**0.0056 ± 0.0006**

Table 3: The placement ladder.

6.2 Ablation status

#	Question it answers	Result
1	Does frame order matter to S2c? (temporal shuffle)	Bit-identical to the unperturbed run at every digit.
2	Is the model reading the sky, or is it reading a plant-level constant?	A stale sky is worse than no sky — vision is read, and read for its timing.

Table 4: Executed ablations and their measured outcomes.

#	Question it answers	Expected behaviour	GPU-h
3	Summariser widened to 4 temporal slices of 4 × 4 blocks — S2c's payload with S2b's fixed-latent queries.	Reliance ≈ 0 ⟶ the mechanism is the forecast-side query.	≈ 24
4	Grid or decoder? S2c with the grid collapsed to 1 × 1	Reliance falls to the S2b level if the grid is what matters; holds if the decoder is.	≈ 24
5	Grid held at 4 × 4, decoder collapsed to one position.	Reliance holds if the grid is the mechanism; falls if the 3-slot decoder is.	≈ 24
6	Is the grid spatially grounded? Frames swapped with another plant's.	NMAE degrades below the vision-free control and reliance turns negative if the gain is plant-specific; indifference would indicate a generic cloudiness prior.	< 1

Table 5: Open ablations, configured but not launched.

6.3 Placement against the baseline suite

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
		This work (MMTSFM placement arms)			
1	**MMTSFM S2c** (ours)	satellite	cross-attention from **future** positions to unpooled memory	**0.5470**	0.1461
3	MMTSFM S2b wide (ours)	satellite	sequence axis, 16 appended tokens (self-attention)	0.5352	0.1484
4	MMTSFM S2b (ours)	satellite	sequence axis, 1 appended token (self-attention)	0.5322	0.1487
5	MMTSFM S2a (ours)	satellite	batch axis, pooled vector (group self-attention)	0.5258	0.1487
7	MMTSFM S1 control (ours)	none	— (vision-free control)	0.5230	0.1506
		Multimodal forecasters (satellite & pseudo-image)			
2	Time-VLM	series **rendered as** images	reuses visual pretraining in the opposite direction	0.5404	—
13	Solar-VLM	satellite + text	vision–language fusion, multi-site	0.4396	0.1514
21	CrossViViT	satellite	cross-attention from history timesteps	0.3491	—
26	Aurora	several	joint multimodal pretraining	0.2324	—
27	SUNSET	sky/satellite	convolutional precedent; joint encoding	0.2162	—
28	UniCast	several	prompting a foundation forecaster	0.1211	—
30	VisionTS++	series **rendered as** images	continual pretraining of a visual backbone	0.0167	—

Table 6: Multimodal leaderboard.

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
		Supervised deep learning			
6	iTransformer + covariates	covariates	channel-inverted self-attention over variates	0.5257	**0.1445**
12	PatchTST	none	—	0.4581	0.1543
14	Temporal Fusion Transformer	covariates	gated residual & temporal self-attention	0.4264	0.1605
15	MLP	none	—	0.4219	0.1624
22	DLinear	none	—	0.3231	0.1746
		Retrieval & frozen-backbone adaptation			
9	TS-RAG	retrieved **numeric** history	concatenated to the context	0.4779	—
10	Cross-RAG	retrieved **numeric** history	cross-attention between query and retrievals	0.4768	—
18	CoRA	covariates	residual adapter on frozen backbone	0.3798	0.1624
		Time-series foundation models (zero-shot & fine-tuned)			
8	Chronos-2, fine-tuned	covariates	group self-attention, fine-tuned	0.5042	0.1494

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
11	Chronos-2, zero-shot	covariates	group self-attention, zero-shot	0.4737	0.1544
20	TTM, fine-tuned	covariates	MLP-Mixer temporal/ channel mixing, fine-tuned	0.3601	0.1716
23	TiRex, zero-shot	none	—	0.2873	0.1826
24	TimesFM, zero-shot	none	—	0.2708	0.1902
32	TTM, zero-shot	covariates	MLP-Mixer, zero-shot	−0.0807	0.2922

Table 7: Unimodal deep learning and foundation model baselines.

Rank	Model	Second modality	Where the fusion sits	Skill score	Ramp NMAE
			Tabular models & classical ML		
16	TabPFN	none	—	0.4063	0.1631
17	LightGBM	covariates	gradient-boosted trees over tabular lags	0.3854	0.1672
19	TabFM ensemble	none	—	0.3626	0.1573
			Reference & statistical baselines		
25	Hourly climatology	none	—	0.2337	0.1665
29	Seasonal naive	none	—	0.1068	0.2056
31	Persistence	none	—	0.0141	0.2550

Table 8: Tabular, classical ML, and reference baselines.